Curriculum Vitae - Sandesh Kalantre

Contact Information

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Interests

Experimental Condensed Matter Physics, Topological Materials, Josephson Junctions
Low-temperature Physics, Nano-fabrication
Quantum Computers, Machine Learning

Education

University of Maryland, College Park July 2018 - Present
Graduate Research Assistant in Physics
JQI-QuICS Lanczos Graduate Fellow
Joint Quantum Institute (JQI)

Indian Institute of Technology, Bombay July 2014 - April 2018
Bachelor of Technology (B.Tech) in Engineering Physics with Honours
Minor in Computer Science
GPA : 9.87/10

Honours and Awards

Institute Silver Medal for graduating with honors at the top of Engineering Physics B.Tech Program, IIT Bombay 2018

Institute Academic Award for highest GPA in the Physics Department, IIT Bombay 2015-2017

Silver Medal at 7th International Olympiad on Astronomy and Astrophysics, Greece 2013

Gold Medal at 6th International Olympiad on Astronomy and Astrophysics, Brazil 2012

Gold Medal at 16th International Astronomy Olympiad, Kazakhstan and a special diploma for **ranking** 1st among participants from 21 countries 2011

Scholarships & Fellowships

JQI-QuICS Lanczos Graduate Fellowship 2018 - Present
Graduate fellowship for working at JQI-QuICS with a $36,000 annual stipend

S. N. Bose Scholars Program 2017
$2000 travel support for summer-research experience at leading US institutions (awarded to the top 50 students from India)

ISTernship - Institute of Science and Technology (IST) Austria 2016
€2700 award to pursue research as part of the ISTernship (awarded to around 40 students worldwide)

National Initiative on Undergraduate Science (NIUS) Program in Physics 2015
Financial support for summer at leading institutions in India (awarded to 20 students in India)

Kishore Vigyan Protsahan Yojana (KVPY) 2013
Financial award for promotion of basic sciences among high school students (awarded to the top 250 students in India)

National Talent Search Scholarship (NTSE) 2010
Awarded by the National Council for Educational Research and Training, Govt. of India to the top 1000 high school students in the country as support for secondary school education.

Publications and Preprints

[1] Justyna P. Zwolak, Sandesh S. Kalantre, Xingyao Wu, Stephen Ragole, and Jacob M. Taylor. "QFlow lite dataset: A machine-learning approach to the charge states in quantum dot experiments". en. In: *PLOS ONE* 13.10 (Oct. 2018), e0205844. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0205844.

[2] Sandesh S. Kalantre, Justyna P. Zwolak, Stephen Ragole, Xingyao Wu, Neil M. Zimmerman, M. D. Stewart, and Jacob M. Taylor. "Machine Learning techniques for state recognition and auto-tuning in quantum dots". In: *arXiv:1712.04914 [quant-ph]* (Dec. 2017). arXiv: 1712.04914. URL: http://arxiv.org/abs/1712.04914.

Research Presentations

[1] *Non-linear dynamics and crisis in RF-driven graphene Josepshon junctions*, APS Mid-Atlantic Meeting, College Park, MD, USA (November 2018)

[2] *ML and Quantum Dots: Generation of Training Datasets and Autotuning*, APS March Meeting, Los Angeles, CA, USA (March 2018)

[3] *ML and Quantum Dots: Navigation in Quantum Dot Land*, Quantum Seminar, NIST, Gaithersburg, MD, USA (July 2017)

[4] *ML and Quantum Dots: Navigation in Quantum Dot Land*, seminar at University of Maryland-College Park, MD, USA (July 2017)

[5] *Spin Qubits in Semiconductors*, Poster for ISTernship Program, IST Austria, Vienna (July 2016)

Key Research Experience

Joint Quantum Institute, University of Maryland, College Park, MD, USA
Graphene Josepshon Junctions
Advisor: Prof. James R. Williams *2018-present*

- Simulations of the Resistively and Capacitively Shunted Junction (RCSJ) Model to understand experimental measurements on a RF-driven graphene Josephson junction
- Explored the non-linear chaotic behaviour of underdamped junctions occuring in simulations and in comparison to the experiment

National Institute of Standards and Technology, Gaithersburg, MD, USA
Machine Learning Techniques for auto-tuning of Quantum Dots
Advisor: Prof. Jacob M. Taylor *Summer 2017*

- Developed novel machine learning techniques with deep and convolutional neural networks to characterize charge configurations of semiconductor quantum dots
- Introduced a procedure of "auto-tuning" to automatically achieve a desired configuration of a quantum dot array

Institute of Science and Technology, Vienna, Austria
Spin Qubits in Semiconductor (Ge) Nanostructures
Advisor: Dr. Giorgos Katsaros *Summer 2016*

- Studied theoretical aspects of transport in quantum dot and nanowire systems in semi-conductors, and their potential application as spin qubits
- Worked on conductance measurements to observe Coulomb blockade and diamonds in p-type transistors at 4 K (Liquid He); Nb superconducting resonators

Indian Institute of Technology, Bombay, India
Quantum Dot Fabrication in Si/SiGe and ESR Selectivity of Donor spin qubits
Advisor: Prof. Suddhasatta Mahapatra *2016-2018*

- Worked in a nano-fabrication facility to pattern nano-scale devices in semiconductor heterostructures

- Worked on a theoretical problem of Electron Dipole Spin Resonance (EDSR) selectivity of Phosphorous Donor qubits in micromagnet fields

Indian Institute of Technology, Bombay, India
Non-equilibrium Green's function calculation of transport in superconducting-normal junctions
Advisor: Prof. Bhaskaran Muralidharan *2016-2018*

- Employed non-equilibrium Green's function formalism to model transport in quantum systems
- Determined the effects of angular momentum subbands and Andreev Bound states to the current-phase relationship in nanowires connected to superconducting contacts

Applied Projects

Lockin amplifier with Arduino *2017*
Built a prototype lock-in amplifier using the I/0 circuitry on a Arduino Uno board

Webcam Spectrograph using Raspberry Pi (RPi) *2016*
Used CD as a diffraction grating & the camera of a RPi to construct a portable spectrograph

Three Body Simulation using FPGA *2016*
Programmed an Altera De0-Nano board using VHDL to perform real time simulation of 3 bodies interacting gravitationally

NumCpp - a mathematical library in C++ *2014*
Wrote a syntax parser and an arbitrary precision mathematical library in C++

Mentoring Experience

Teaching Assistant
For Quantum Physics, Electromagnetism, Partial Differential equations courses *2015-2017*

- Mentored a batch of around 50 students in the course content and involved in the evaluation of exams and quizzes

Institute Student Mentor *Indian Institute of Technology, Bombay* *2017*

- Mentor to a wing of 12 freshmen to assist them in settling to college life and the academic setting of an university
- Involved in conducting basic computer learning classes for the computer illiterate students to help them cope with computer programming courses

Resource Person and Student Facilitator
Indian Astronomy Olympiad Program, HBCSE - TIFR, Mumbai *2015*

- Mentor to about top 30 high school students at the selection camp for the Indian Team to the International Olympiad on Astronomy and Astrophysics (IOAA)
- The team was awarded 3 Gold & 2 Silver medals in the International Olympiad on Astronomy and Astrophysics (IOAA), Indonesia, 2015, which was *India's best result in 9 years.*

Positions of Responsibility

General Secretary *Department of Physics, IIT Bombay* *2017-2018*

- As the top student representative of the physics department's bachelor and masters' students, I co-ordinate and communicate students issues to the faculty and other representatives at the institute level.
- Responsible for the organization of departmental events such as the valedictory function, degree convocation and freshmen introduction sessions.

Manager - Maths and Physics Club, IIT Bombay *2016-2017*
Students' Technical Activities Body (STAB), IIT Bombay

- Led a team of 6 sophomore conveners in organization of events fostering to the enthusiasm of students in Physics and Mathematics; catering to around **400-500** students on campus and having an outreach of around **6000** online
- Introduced a new initiative in summer 2016, *Summer of Science* in which students studied varied topics such as Game Theory, Cosmology and Quantum Computation under senior student mentors

Academic Resource Person & IPho-rum developer *2015*
Browser Application for International Physics Olympiad (IPhO), Mumbai, 2015

- Worked in a team of 2 in development of IPho-rum — a browser based application for tasks such as voting, translation upload and feedback submission among *approx 100 users*; used *Node.js* as the server application and Mongodb as the primary database
- Involved as a student grader in the academic team

Popular Talks

A Brief Introduction to Quantum Computation
Maths and Physics Club, IIT Bombay *March 2018*
Starting with the history of computing, I presented how quantum computing emerges as the natural 'next step'. I talked about killer-apps such as integer factorization, quantum many-body simulation.

Cosmology: Dark Things on a White Background
Kritikka - The Astronomy Club, IIT Bombay *March 2018*
Gave an elementary introduction to cosmological models using Newtonian mechanics including the necessity of dark matter and dark energy to explain observations.

Celestial Mechanics
Kritikka - The Astronomy Club, IIT Bombay *September 2015,2016*
Presented a complete derivation of the dynamics of the two-body problem using Newtonian mechanics. Talked about chaos in dynamics systems in relation on the three and N body problems.

Technical Skills

Programming : C++, Python

Scientific Packages : TensorFlow, Synopsys - TCAD, NumPy, SciPy, Matplotlib, QuTip, Astropy, gnuplot, MATLAB, Mathematica, Octave, Origin, GNU MPFR library

Softwares : LaTeX, git, InkScape, node.js, Wireshark, PyGTK

Extra-curricular activities

- Won a Bronze medal in Astronomy hackathon as part of InterIIT Tech Meet, IIT Madras, December 2017.
- Interested in all things in print, from abstract mathematics, history & mythology to Shakespeare & Whitman
- Built a Kelvin Water Dropper and demonstrated build-up of a potential difference under Maths and Physics club, IIT Bombay
- Secured first place in *Online Physics Brawl 2016* in the open category
- Associated with National Service Scheme as a volunteer at NGOs for underprivileged children in Powai, Mumbai
- Secured first prize in Debate competition held at Dyanamata High School, Maharashtra

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